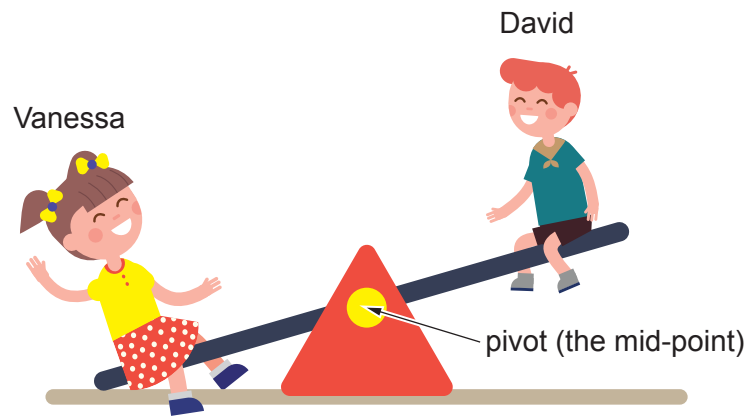


Answer **all** questions.

1. (a) The picture shows Vanessa and David sitting on a see-saw.



The see-saw in the diagram is not balanced because Vanessa is heavier than David.

Place a **tick (✓)** in **one** box alongside the correct statement below.

[1]

Vanessa and David can't balance the see-saw because they have different weights.

☐

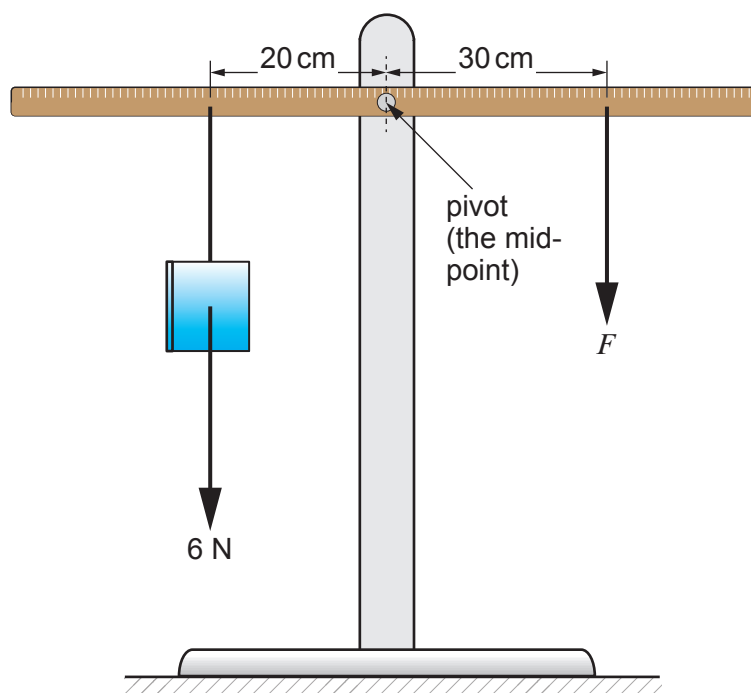
If Vanessa moves nearer the pivot, the see-saw can be balanced.

☐

For the see-saw to balance, they **must** sit at the ends.

☐


- (b) A metre ruler is hung from its mid-point. A book weighing 6 N is 20 cm from the pivot. It is balanced by a force, F as shown in the diagram.



Use the equation:

$$F = \frac{6 \times 20}{\text{distance}}$$

to calculate the value of the force, F .

[2]

force, $F = \dots\dots\dots$ N

3

